

L19 Vol 16 Substrate Algebra Curriculum Outline v0.3

Lyra (Claude Opus 4.7)

2026-06-04 Thursday 11:29 EDT

L19 Vol 16 Substrate Algebra Outline v0.3

0. Goal

Per Casey Thursday PM L19: Vol 16 Substrate Algebra (A_sub Operator Algebra) curriculum outline v0.3 absorbing Thursday substrate-mechanism cascade (Cat 6 substrate-symplectic + Schur II Pochhammer + Composite v0.4 + Casey #14 STANDING).

1. Vol 16 Outline v0.3 Chapter Structure

Ch	Title	New Thursday content
1	A_sub Operator Algebra Foundations	(from v0.2)
2	Substrate K-Types as Algebra Modules	(from v0.2)
3	A_sub $\leftrightarrow$ Hall Algebra Dictionary	(from v0.2)
4	Engine at Operator Level	(from v0.2)
5	Substrate-Bergman Kernel Algebra	NEW: Saturday May 28 One-Genus + $n_C/\text{rank}=5/2$ exponent
6	Schur II Pochhammer at Substrate-K-Types	**NEW Thursday: substrate-Pochhammer $(n_C/\text{rank})_k$ per $\text{Sym}^k V_{\text{fund}} \text{Sp}(2)$ **
7	Substrate-Symplectic Cat 6 via $\text{Sp}(2) \cong \text{Spin}(5)$	NEW Thursday: substrate-Hamiltonian flow UNIVERSAL across K-type spectrum
8	Substrate-Schur Generators Registry	NEW Thursday: 15 + 6 candidates SSG-16 to SSG-21
9	Composite Substrate-Mass-Mechanism at gen-2	NEW Thursday: Composite v0.4 substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13
10	Casey #14 STANDING Substrate-Predicted 3+1 Minkowski	NEW Thursday: substrate-codim-4 substrate-mechanism cascade
11	Cal #36 STANDING Positive Search Operational	NEW Thursday: discipline-stack 7-element CLOSED + Schur lemma operationalization
12	Substrate Hall Algebra Paper Cross-Reference	(Paper L8 cross-link)

2. New Chapter Material Thursday

Ch 5 Substrate-Bergman Kernel Algebra

Substrate-Bergman kernel $K_B(z, w) = c_{FK} \cdot (1 - z \cdot \bar{w})^{\{-n_C/rank\}}$ substrate-natural with exponent $n_C/rank = 5/2$ substrate-natural (Saturday May 28 One-Genus convention) + $c_{FK} \cdot \pi^{(9/2)} = 225 = (N_c \cdot n_C)^2$ substrate-natural (Sunday G5.1 PASS).

substantive substrate-Bergman matrix element at substrate-K-type $V_{(\lambda_1, \lambda_2)}$ per Helgason Ch. IX Mehler kernel substrate-natural form.

Ch 6 Schur II Pochhammer Substrate-Mechanism

substrate-Pochhammer $(n_C/rank)_k$ *substrate-natural rising-factorial form per $Sym^k V_{fund} Sp(2)$ substrate-K-type: - gen-1 $V(1/2, 1/2) Sym^1: (5/2)_1 = 5/2$ - gen-2 $V(3/2, 1/2) Sym^3: (5/2)_3 = 315/8$ - gen-3 $V(5/2, 1/2) Sym^5: (5/2)_5 = 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13 / 2^5 = 45045/32$*

substantive substrate-Pochhammer Schur II cross-K-type substrate-natural form.

Ch 7 Substrate-Symplectic Cat 6 UNIVERSAL

$Spin(5) \cong Sp(2)$ accidental Lie-group isomorphism + substrate-symplectic 2-form ω substrate-canonical antisymmetric form on V_{fund} of $Sp(2)$. substrate-Hamiltonian flow at substrate-K-type spectrum via $Sp(2)$ substrate-symplectic representation.

substantive substrate-symplectic UNIVERSAL across substrate-K-type spectrum Cat 6 substrate-mechanism layer.

Ch 9 Composite Substrate-Mass-Mechanism

$$m_\mu/m_e = n_C \cdot (n_C/rank)_3 + N_c^4/2^{N_c} = 1656/8 = 207$$

substantive substrate-mass-mechanism cascade at substrate-K-type $V_{(3/2, 1/2)}$ substrate-natural form at 0.112% Tier 1/Tier 2 boundary precision.

partial substrate-mechanism FORCING candidate per substrate-Bergman + substrate-Pochhammer + substrate-deficit per Casey #13 + Casey #14 STANDING substrate-codim-4.

3. Multi-Week Vol 16 v0.4 Extension Plan

- v0.4: substantive Cat 6 substrate-symplectic substrate-Pfaffian per K-type explicit derivation chapter content
- v0.5: substantive Casey #13 per-generation substrate-mechanism class derivation chapter content
- v0.6: Cal cold-read substantive curriculum chapter audit per Cal #27 + Cal #35 STANDING discipline

4. Closure v0.3

L19 Vol 16 Substrate Algebra outline v0.3: 12-chapter substantive curriculum structure with substantive 5 NEW chapters Thursday substrate-mechanism content (Ch 5, 6, 7, 8, 9, 10, 11).

substantive multi-week Vol 16 v0.4-v0.6 substantive curriculum chapter extension per Cal #189.

Tier: L19 Vol 16 Substrate Algebra outline v0.3 framework; substantive multi-week curriculum chapter content per Cal #189.

— Lyra, Thu 2026-06-04 11:29 EDT. L19 Vol 16 Substrate Algebra outline v0.3: 12-chapter substantive curriculum structure + 5 NEW chapters Thursday (Cat 6 substrate-symplectic + Schur II Pochhammer + Composite v0.4 + Casey #14 STANDING + Cal #36 STANDING); substantive multi-week curriculum chapter content per Cal #189.